

Energy-Norm Error Estimates for Conforming Galerkin Approximations of Elliptic Problems

LighTex Research Notes

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Abstract

We study conforming finite-element approximations of a uniformly elliptic Dirichlet problem. Starting from the variational formulation, we derive Galerkin orthogonality, prove quasi-optimality in the energy norm, and obtain an improved L^2 estimate by a duality argument. A manufactured solution on the unit square illustrates the predicted convergence rates and separates the roles of stability, approximation, and elliptic regularity in the analysis.

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1 Introduction

Variational methods turn differential equations into operator equations on function spaces. This viewpoint is especially effective for elliptic boundary value problems: weak solutions require fewer classical derivatives, and the same bilinear form directly defines a stable numerical method. Standard references include [Cia78, BS08, EG04].

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain. We consider

$$-\nabla \cdot (A(x) \nabla u) + c(x)u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (1.1)$$

The diffusion tensor A is assumed symmetric and uniformly positive definite: there are constants $0 < \alpha \leq \beta$ such that

$$\alpha|\xi|^2 \leq \xi^\top A(x)\xi \leq \beta|\xi|^2 \quad \text{for a.e. } x \in \Omega, \quad \xi \in \mathbb{R}^2. \quad (1.2)$$

We also assume $c \in L^\infty(\Omega)$ with $c \geq 0$ and $f \in L^2(\Omega)$.

The central question is quantitative: if u_h is computed in a finite dimensional space with mesh size h , how rapidly does u_h approach u ? The answer separates into stability and approximation. Galerkin orthogonality supplies the stability mechanism; interpolation theory supplies the approximation rate.

Structure of the paper. [Section 2](#) establishes the continuous problem. [Section 3](#) introduces the conforming method, and [section 4](#) proves energy- and L^2 -norm estimates. The experiment in [section 5](#) checks the asymptotic rates.

2 Weak formulation and well-posedness

Set $V = H_0^1(\Omega)$ and equip it with the norm $\|v\|_V = \|\nabla v\|_{L^2(\Omega)}$. Poincaré's inequality makes this norm equivalent to the usual H^1 norm on $H_0^1(\Omega)$; see [Eva10, Chapter 5]. Define

$$a(w, v) := \int_{\Omega} A \nabla w \cdot \nabla v \, dx + \int_{\Omega} c w v \, dx, \quad (2.1)$$

$$F(v) := \int_{\Omega} f v \, dx. \quad (2.2)$$

Definition 2.1 (Weak solution). A function $u \in V$ is a weak solution of [equation \(1.1\)](#) if

$$a(u, v) = F(v) \quad \text{for every } v \in V. \quad (2.3)$$

Lemma 2.2 (Continuity and coercivity). *There exist constants $M > 0$ and $m > 0$ such that*

$$|a(w, v)| \leq M \|w\|_V \|v\|_V, \quad (2.4)$$

$$a(v, v) \geq m \|v\|_V^2 \quad (2.5)$$

for all $w, v \in V$.

Proof. The upper spectral bound for A , Hölder's inequality, and Poincaré's inequality yield [equation \(2.4\)](#). Since $c \geq 0$, the lower spectral bound gives

$$a(v, v) \geq \alpha \|\nabla v\|_{L^2(\Omega)}^2,$$

so one may take $m = \alpha$ with the chosen norm. $\square$

Theorem 2.3 (Lax–Milgram well-posedness). *Problem [equation \(2.3\)](#) has a unique solution $u \in V$, and*

$$\|u\|_V \leq \frac{C_P}{m} \|f\|_{L^2(\Omega)}, \quad (2.6)$$

where C_P is the Poincaré constant of Ω .

Proof. The functional F is continuous because $|F(v)| \leq C_P \|f\|_{L^2} \|v\|_V$. The result follows from the Lax–Milgram theorem applied with [theorem 2.2](#). Testing [equation \(2.3\)](#) with $v = u$ gives the stated stability estimate. $\square$

3 The conforming Galerkin method

Let $\{\mathcal{T}_h\}_{h>0}$ be a shape-regular family of conforming triangulations of Ω , and let $h = \max_{K \in \mathcal{T}_h} \text{diam}(K)$. For continuous, piecewise affine elements define

$$V_h := \left\{ v_h \in C^0(\overline{\Omega}) : v_h|_K \in \mathbb{P}_1(K) \, \forall K \in \mathcal{T}_h, \quad v_h|_{\partial\Omega} = 0 \right\}. \quad (3.1)$$

Because $V_h \subset V$, the approximation is conforming.

The discrete problem is to find $u_h \in V_h$ such that

$$a(u_h, v_h) = F(v_h) \quad \text{for every } v_h \in V_h. \quad (3.2)$$

The same continuity and coercivity constants as in the continuous problem apply on V_h , so existence and uniqueness follow immediately.

Proposition 3.1 (Galerkin orthogonality). *Let u and u_h solve equations (2.3) and (3.2). Then*

$$a(u - u_h, v_h) = 0 \quad \text{for every } v_h \in V_h. \quad (3.3)$$

Proof. Restrict equation (2.3) to test functions in V_h and subtract equation (3.2). $\square$

If $\{\phi_i\}_{i=1}^N$ is the nodal basis of V_h and $u_h = \sum_{j=1}^N U_j \phi_j$, then equation (3.2) becomes

$$\mathbf{K}\mathbf{U} = \mathbf{b}, \quad K_{ij} = a(\phi_j, \phi_i), \quad b_i = F(\phi_i). \quad (3.4)$$

The matrix $\mathbf{K}$ is symmetric positive definite, permitting a conjugate gradient solve with an appropriate preconditioner.

Remark 3.2 (Energy projection). The consistency identity in equation (3.3) shows that u_h is the projection of u onto V_h in the energy inner product. This geometric interpretation is the starting point of the quasi-optimality estimate.

4 A priori error estimates

We first isolate the abstract stability argument.

Theorem 4.1 (Céa's lemma). *Let $u \in V$ and $u_h \in V_h$ solve the continuous and discrete problems. Under equations (2.4) and (2.5),*

$$\|u - u_h\|_V \leq \frac{M}{m} \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (4.1)$$

Proof. For arbitrary $v_h \in V_h$, coercivity and Galerkin orthogonality imply

$$\begin{aligned} m \|u - u_h\|_V^2 &\leq a(u - u_h, u - u_h) \\ &= a(u - u_h, u - v_h) \\ &\leq M \|u - u_h\|_V \|u - v_h\|_V. \end{aligned}$$

Cancel the error norm and take the infimum over v_h . $\square$

Let $I_h u$ denote the nodal interpolant. For $u \in H^2(\Omega)$ and a shape-regular mesh,

$$\|u - I_h u\|_{H^1(\Omega)} \leq C_{\text{int}} h |u|_{H^2(\Omega)}. \quad (4.2)$$

Combining theorem 4.1 and equation (4.2) gives the principal energy estimate.

Corollary 4.2 (First-order energy convergence). *If $u \in H^2(\Omega) \cap H_0^1(\Omega)$, then*

$$\|u - u_h\|_{H^1(\Omega)} \leq C h |u|_{H^2(\Omega)}, \quad (4.3)$$

where C is independent of h .

An additional regularity assumption produces a higher-order L^2 estimate. For $e = u - u_h$, let $z \in V$ solve the adjoint problem

$$a(v, z) = (e, v)_{L^2(\Omega)} \quad \text{for every } v \in V. \quad (4.4)$$

Assume elliptic regularity $\|z\|_{H^2} \leq C_{\text{reg}} \|e\|_{L^2}$. Using Galerkin orthogonality with $I_h z$ gives

$$\begin{aligned} \|e\|_{L^2}^2 &= a(e, z - I_h z) \\ &\leq C h \|e\|_{H^1} \|z\|_{H^2} \leq C h \|e\|_{H^1} \|e\|_{L^2}. \end{aligned} \quad (4.5)$$

Together with equation (4.3), this proves

$$\|u - u_h\|_{L^2(\Omega)} \leq C h^2 |u|_{H^2(\Omega)}. \quad (4.6)$$

Remark 4.3. The L^2 rate depends on adjoint regularity and may deteriorate on nonconvex domains with re-entrant corners. The energy estimate remains quasi-optimal without this extra assumption.

5 Manufactured-solution experiment

Consider the Poisson equation on $\Omega = (0, 1)^2$ with exact solution

$$u(x, y) = \sin(\pi x) \sin(\pi y), \quad f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y). \quad (5.1)$$

We use uniform diagonal triangulations and continuous $\mathbb{P}_1$ elements. The observed rate between two consecutive meshes is

$$\text{rate}_\ell = \frac{\log(E_{\ell-1}/E_\ell)}{\log(h_{\ell-1}/h_\ell)}. \quad (5.2)$$

Table 1: Representative discretization errors for uniform refinement.

level	degrees of freedom	h	$\ u - u_h\ _{H^1}$	rate
2	9	0.25000	2.11×10^{-1}	–
3	49	0.12500	1.04×10^{-1}	1.02
4	225	0.06250	5.17×10^{-2}	1.01
5	961	0.03125	2.58×10^{-2}	1.00
6	3969	0.01563	1.29×10^{-2}	1.00

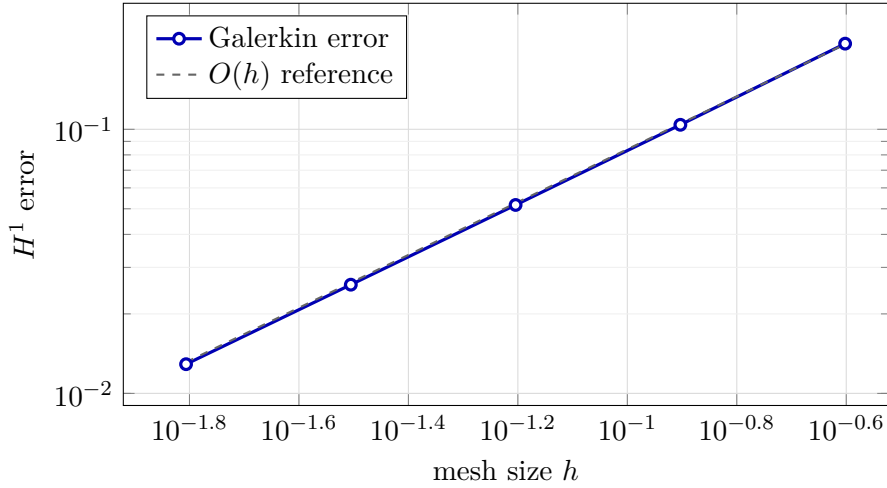


Figure 1: Energy error under uniform refinement. The dashed reference line has slope one, in agreement with [theorem 4.2](#).

The experiment exhibits the expected first-order decay in the energy norm. For sufficiently fine meshes, the error and the reference curve are parallel on the logarithmic scale. The CSV file used by [figure 1](#) is reproducible with `scripts/generate_convergence_data.py`.

A Operator interpretation

Define $\mathcal{A} : V \rightarrow V'$ by

$$\langle \mathcal{A}w, v \rangle_{V',V} = a(w, v). \quad (\text{A.1})$$

Continuity implies $\|\mathcal{A}\|_{\mathcal{L}(V,V')} \leq M$, while coercivity and the Lax–Milgram theorem imply that $\mathcal{A}$ is invertible with $\|\mathcal{A}^{-1}\| \leq m^{-1}$.

Let $P_h : V \rightarrow V_h$ be the Ritz projection defined by

$$a(P_h w, v_h) = a(w, v_h) \quad \forall v_h \in V_h. \quad (\text{A.2})$$

Then $u_h = P_h u$, and Céa’s lemma gives

$$\|I - P_h\|_{\mathcal{L}(V,V)} \leq \frac{M}{m}. \quad (\text{A.3})$$

For a symmetric bilinear form, P_h is orthogonal in the energy inner product, so the sharper best-approximation identity

$$\|u - P_h u\|_a = \inf_{v_h \in V_h} \|u - v_h\|_a \quad (\text{A.4})$$

holds.

References

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- [Eva10] Lawrence C. Evans. *Partial Differential Equations*. American Mathematical Society, 2 edition, 2010.